



Agent-like Behavior in Single LLMs

Can a single chatbot do real biomedical research? A live demo with synthetic patient data.

RSE

Capabilities and Limitations

What a single LLM can and cannot do in a biomedical research context

Capabilities

- Can analyze uploaded data and answer follow-up questions
- Can generate code, run reasoning steps, and revise conclusions
- Can detect simple statistical pitfalls, such as confounding

Limitations

- Limited context over longer workflows
- Cannot reliably handle large datasets or direct hardware access
- Weak persistent memory and no true agent-to-agent coordination

SYSTEM PROMPT USED FOR THE DEMO

You are a statistical assistant for biomedical research. You work in R. After a dataset is uploaded, you briefly summarize it and then wait for questions from the physician/researcher. Your responses will contain a single chunk of R code that can be copy-pasted, along with a brief commentary on the results.

The Dataset

1,000 synthetic patient records with 7 variables and a built-in confounder

- Variables include **sex, age, episodes, infarction, antihypertensives, naive CD4**
- Roughly balanced by sex, no missing values
- Built-in patterns: **naive CD4 decreases with age**, antihypertensive use increases with age, infarction more frequent in men
- Main question: **Do antihypertensives affect naive CD4 T-cell levels?**
- A naive comparison suggests an effect, but **age is the key confounder**

You are a statistical assistant for biomedical research. You work in R. After a dataset is uploaded, you briefly summarize it and then wait for questions from the physician/researcher. Your responses will contain a single chunk of R code that can be copy-pasted, along with a brief commentary on the results.

Thought for 5s

Understood.

Once you upload a dataset, I will:

- briefly summarize its structure and key variables,
- then wait for your specific research/statistical question,
- and answer with exactly one copy-pasteable R code chunk plus a brief interpretation.

Upload the data when ready.

Data Upload + Summary

The LLM inspects the uploaded table, identifies variable types, and reports a clean summary

I'll inspect the uploaded table and give you a short data summary first.

Thought for 20s >

 R



```
df <- read.csv("data130425.csv", sep = "\t")

str(df)
summary(df)
colSums(is.na(df))
table(df$sex, useNA = "ifany")
table(df$episodes, useNA = "ifany")
table(df$infarction, useNA = "ifany")
table(df$antihypertensives, useNA = "ifany")
```

Dataset summary: 1000 rows and 7 variables. The variables appear to be `pac` (ID), `sex`, `episodes`, `infarction`, `antihypertensives`, `naive_cd4`, and `age`. The table looks clean at first glance, with no obvious missing values in the uploaded file. `sex` is  orical; the others are numeric, with `episodes`, `infarction`, and `antihypertensives` likely binary indicators. Ask your first research question and I'll continue with one R chunk plus a short interpretation.

Research Question + Initial Analysis

User asks about antihypertensives and CD4 levels — the model runs crude + adjusted analyses

How does administration of antihypertensive drugs influence the levels of naive CD4 T cells?

I'm checking the relevant variables and will give you a single R chunk for the comparison.

Thought for 25s >

```
<> R
```

```
# crude comparison
aggregate(naive_cd4 ~ antihypertensives, data = df, function(x) {
  c(n = length(x), mean = mean(x), sd = sd(x), median = median(x))
})

t.test(naive_cd4 ~ antihypertensives, data = df)

# adjusted analysis
fit <- lm(naive_cd4 ~ antihypertensives + age + sex, data = df)
summary(fit)
confint(fit)
```

Crude comparison suggests that patients receiving antihypertensives have lower naive CD4 levels on average than untreated patients. But after adjustment for age and sex, the coefficient for `antihypertensives` is small and not statistically significant, while `age` shows a clear negative association with naive CD4. In this dataset, the apparent unadjusted difference is therefore most likely explained mainly by age rather than by antihypertensive treatment itself.

Detailed Results + Interpretation

After adjustment for age and sex, the apparent treatment effect vanishes — age is the confounder

include detailed results (including p-values in your response)

I'm extracting the actual estimates and p-values so I can report them concretely, not just generically.

Thought for 28s >

```
R

# Descriptive statistics by treatment group
aggregate(naive_cd4 ~ antihypertensives, data = df, function(x) {
  c(n = length(x),
    mean = mean(x),
    sd = sd(x),
    median = median(x))
})

# Unadjusted comparison
t.test(naive_cd4 ~ antihypertensives, data = df)

# Adjusted model
fit <- lm(naive_cd4 ~ antihypertensives + age + sex, data = df)
summary(fit)
confint(fit)
```

Detailed results: untreated patients (`antihypertensives = 0`) had mean naive CD4 of about **69.5** (SD 13.4, n = 566), while treated patients (`antihypertensives = 1`) had mean **66.3** (SD 13.0, n = 434). The unadjusted difference was therefore about **3.2 units lower** in treated patients, and this was statistically significant by Welch t-test (p = 0.00013).

After adjustment for `age` and `sex` in a linear model, the effect of antihypertensive treatment was no longer statistically significant: coefficient for `antihypertensives` about **+0.51**, p = **0.498**, with 95% CI approximately **-0.97 to 1.98**. By contrast, `age` was strongly associated with lower naive CD4: coefficient about **-0.38 per year**, p < 0.001. `sex` was not significant (p = 0.654).

Interpretation: in this dataset, antihypertensive treatment appears associated with lower naive CD4 only in the crude analysis; after accounting for age and sex, there is **no evidence of an independent treatment effect**. The apparent crude difference is mainly explained by confounding, especially by age.